Farah BEN SLIMENE

Robotics Engineering Student – Perception, Control and Learning



6-month end-of-studies internship from March 2027
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PROFILE

Final-year engineering student on a **double degree** in **Mechatronics** (ENISo, Tunisia) and **Innovative Mechanisms and Robots** (**INP SIGMA Clermont**, France), looking for a **6-month end-of-studies internship in robotics**. I work across **perception**, **control** and **learning**: **deep reinforcement learning** (**PPO**) for the **whole-body control** of a mobile manipulator, a **ROS 2** vision-to-motion pipeline on a **real UR3e arm**, and **SLAM** and autonomous navigation with **Nav2**. **Python**, **C++**, **PyTorch**, **Git**.

EDUCATION

Engineering Degree – Innovative Mechanisms and Robots (IMR) *09/2025 – 09/2027*
INP SIGMA Clermont, Clermont-Ferrand, France (double degree) – coursework includes rigid-body dynamics
National Engineering Degree in Mechatronics *09/2023 – 09/2027*
National Engineering School of Sousse (ENISo), Tunisia
Preparatory Classes in Mathematics and Physics (MP) *09/2021 – 06/2023*
Preparatory Institute for Engineering Studies of Bizerte, Tunisia – national entrance exam rank: 494/1402

PROFESSIONAL EXPERIENCE

Research Intern – Deep Reinforcement Learning for Robot Control 05/2026 – 09/2026
CIAD Laboratory, University of Technology of Belfort-Montbéliard (UTBM), Montbéliard, France

- Worked on the **whole-body control** of a mobile manipulator (AgileX Scout 2.0 base, Doosan 6-axis arm) in a **Unity ML-Agents** simulation.
- Trained control policies with **PPO (Proximal Policy Optimization)** in **Stable-Baselines3**.
- Built and trained two **Deep Reinforcement Learning** control architectures:
  - **Hybrid model**: a model-based **PBVS** (Position-Based Visual Servoing) + **WDLS** (Weighted Damped Least Squares) controller, provided by the lab and ported to Unity (C#) by me, drives the robot; the policy tunes its weights to decide how much the base and the arm should move.
  - **End-to-end model**: the policy directly outputs the 8 velocities of the robot (base and arm).
- Compared both models on accuracy, time to reach the target and trajectory smoothness.

Engineering Intern – Automation and Machine Vision 07/2025 – 08/2025
CASCO Automotive, Tunisia

- Deployed real-time **machine-vision quality control** (Keyence camera) in an industrial environment.
- Programmed an automated assembly station for high-voltage components (frunk).

PROJECTS

3D Perception and Bin-Picking with a UR3e Arm 2025 – 2026
*UR3e, RealSense D435, ROS 2, YOLOv11, MoveIt 2 – European project **REMAIN**, team of 6*

- Built a complete **ROS 2** pipeline from perception to execution, with an eye-in-hand **RGB-D camera** and **hand-eye calibration**.
- Trained a **YOLOv11 instance segmentation** model (1,200 hand-labelled images) and computed the **grasp pose** from the masks (PCA).
- Planned collision-aware **pick-and-place** trajectories with **MoveIt 2** and ran them on a real **UR3e** arm.

Reinforcement Learning for Robotic Navigation 2025 – 2026
TurtleBot3, ROS 2 / Gazebo, PyTorch – Team project

- Implemented **Policy Gradient**, **DQN** and **Actor-Critic** from scratch in **PyTorch** for **LiDAR**-based obstacle avoidance.
- Built a **curriculum-learning DQN** with a distributed **ROS 2** service architecture.

Autonomous Mobile Robot: SLAM and Navigation 12/2025
ROS 2 Humble, Gazebo, slam_toolbox, Nav2 – Academic project

- Modelled a differential-drive robot (URDF/xacro, **ros2_control**) with a 360° **LiDAR** and a camera, in a custom Gazebo maze.
- Mapped the maze with **slam_toolbox**, localised on the saved map with **AMCL**, and navigated to goal poses with **Nav2** (costmaps, planner, controller).

TECHNICAL SKILLS

Perception: computer vision, instance segmentation (YOLO), grasp pose estimation, hand-eye calibration, OpenCV, LiDAR
Robotics: ROS 2 (Humble), MoveIt 2, Nav2, SLAM (slam_toolbox), AMCL, kinematics, TF2, visual servoing
Machine Learning: PyTorch, Keras, deep learning (CNN, RNN/LSTM), deep RL (PPO, DQN, Actor-Critic)
Simulation and CAD: Gazebo, Unity, Webots, MATLAB/Simulink, MSC Adams, SolidWorks (CSWP)
Programming and tools: Python (NumPy, SciPy), C++, C#, Git/GitHub

LANGUAGES

English: B2 (TOEIC 920/990) French: B2 Arabic: native Turkish: A2